

Enrolment No: S24CSEU2299

Name of Student: Rishabh



BENNETT
UNIVERSITY
THE TIMES GROUP

NAAC
GRADE A+
ACCREDITED UNIVERSITY

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Even Semester 2025-26

B.Tech. (CSE) – 4th Semester

CSET 209 - Operating Systems

Time: 2hrs.

Max Marks: 40

Note: Student must attempt 5 Questions in total. First question is compulsory and student must attempt question(s) from each section. Marks are indicated against each question.

PART-A

Q.1 Attempt any six questions out of the following:

2*6=12 Marks

- (a) Critically examine how the key features of an operating system contribute to effective process management and resource utilization. [CO-1] [L-4] 2
- (b) Critically analyze the role of system calls as an interface between user programs and the operating system, with particular emphasis on their contribution to resource management. [CO-1] [L-4] 2
- (c) Apply your understanding of memory management to analyze how an operating system allocates and reclaims memory for processes during execution. [CO-2] [L-3] 2
- (d) Using your understanding of memory management, describe how demand paging works and how it optimizes memory usage. [CO-2] [L-2] 2
- (e) Apply your knowledge of cloud computing to analyze how it supports the storage and retrieval of data and applications over the internet. [CO-3] [L-3] 2
- (f) Using your understanding of operating systems, explain how distributed computing allows computers to share resources and work together efficiently. [CO-3] [L-2] 2
- (g) Explain a software-based solution to the Critical Section Problem (CSP) and how it helps prevent concurrent access issues in an operating system. [CO-2] [L-2] 2
- (h) Apply your understanding of the Readers–Writers Problem to suggest a method that enables safe access to shared data while avoiding deadlock. [CO-2] [L-3] 2

PART-B

Note: Attempt any two questions out of the following:

- Q.2**
 - a) Explain the role of the kernel in managing processes in an operating system. [3]
 - b) Apply your understanding of the Process Control Block (PCB) to describe how the kernel uses it to switch between two running processes. [3][CO-1] [L-3] 6
- Q.3** A system has 4-page frames, and the page reference string is: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. [CO-2][L-4] 6
 - a) Using the Optimal page replacement algorithm, show how the contents of the frames change after each page reference. [4]
 - b) Calculate and analyze the total number of page faults that occur during this sequence. [2]
- Q.4** A company wants to consolidate its servers by running multiple operating systems on a single physical machine using virtualization. [CO-3] [L-3] 6

- a) Explain how virtualization allows multiple operating systems to share the same hardware resources. [3]
- b) Apply your understanding of virtualization to suggest two practical benefits this approach provides in terms of resource utilization and system efficiency. [3]

6*2=12 Marks

PART-C

Q.5 A system has five processes with the following arrival and burst times. [CO-2] [L-4] 8

Process	Arrival Time (ms)	Burst Time (ms)
P1	0	10
P2	1	5
P3	2	8
P4	3	6
P5	4	2
P6	5	4

The CPU uses Round Robin Scheduling with a time quantum of 4 ms.

- a) Draw the Gantt chart for the above processes. [2]
- b) Calculate the waiting time and turnaround time for each process. [3]
- c) Compute the average waiting time and average turnaround time. [3]

OR

A disk has request queues at the following cylinder numbers: 120, 45, 75, 10, 150, 30, 90, 60. The disk head is currently at cylinder 40, and the disk has cylinders numbered from 0 to 199. [CO-2] [L-3] 8

- a) Using the C-SCAN (Circular SCAN) disk scheduling algorithm, determine the order in which the requests are serviced. [4]
- b) Calculate the total head movement for servicing all the requests using C-SCAN. [4]

Q.6 A system has 6 processes (P0 to P5) and 3 resource types (A, B, C). The maximum demand, allocation, and available resources are given below: [CO-2] [L-3] 8

Process	Max (A, B, C)	Allocation (A, B, C)
P0	7, 5, 3	0, 1, 0
P1	3, 2, 2	2, 0, 0
P2	9, 0, 2	3, 0, 2
P3	2, 2, 2	2, 1, 1
P4	4, 3, 3	0, 0, 2
P5	5, 4, 3	1, 2, 1

Available resources: A = 3, B = 2, C = 2

- a) Calculate the Need matrix for all processes. [2]
- b) Using the Banker's Algorithm, determine whether the system is in a safe state. If yes, find a safe sequence. [6]

OR

A system has 4 page frames and the page reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5. [CO-2] [L-3] 8

- a) Using the FIFO (First-In, First-Out) page replacement algorithm, show the pages present in each frame after every page reference. [2]
- b) Mark where page faults occurred and determine the total number of page faults. [2]
- c) Explain what thrashing is, and discuss whether it could occur if the system had only 3 page frames. [4]

8*2=16 Marks
